How to Steal Oblivious Transfer from Minicrypt

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Abstract

The celebrated work of Impagliazzo and Rudich (STOC'89) provides an oracle separation between those primitives implied by a random oracle (RO) and those that imply key agreement and public-key cryptography. For the last 36 years, this result seemed to cleanly separate two worlds: Minicrypt, which is often described as what can be achieved from only ROs, and Cryptomania, which is a world where public-key cryptography exists.

This work presents a natural primitive, called an oblivious interactive hash function (OIHF), and shows the following:

(1) OIHFs can be constructed from ROs.

(2) OIHFs can be constructed from oblivious transfer (OT), and hence they are implied by various well-studied public-key-style assumptions.

(3) The existence of an OIHF implies OT, via a non-black-box reduction.

Point (1) places the primitive into Minicrypt, point (2) implies that numerous well-studied standard-model complexity assumptions imply that OIHFs exist, and point (3) shows that this primitive circumvents the barrier imposed by Impagliazzo and Rudich.

Our results show that a protocol constructible in Minicrypt implies OT. However, this *does not* imply that OT can be constructed from RO; indeed, Impagliazzo/Rudich shows this is impossible. Our non-black-box reduction is well-defined only for standard-model OIHFs, i.e. ones that do not use RO, and at present we can only construct standard-model OIHF in Cryptomania. This is to say that we do not propose an OT protocol that can be constructed from a cryptographic hash function. Rather, our results call into question the value of the random oracle model as an idealization of Minicrypt, and they question where the boundaries of Minicrypt lie.

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1 Introduction

In their seminal work [IR89], Impagliazzo and Rudich demonstrated an oracle separation between a random oracle (RO) and key exchange. Their work helped to establish a widely-adopted framework for categorizing cryptographic primitives into a symmetric-key "world" (Minicrypt) and a public-key world (Cryptomania) [Imp95]. Although [IR89] explained that their separation could plausibly be circumvented by using Minicrypt primitives in a non-black-box manner, to our knowledge, no prior work showed how, and the above framework remained unambiguous.

We leverage a non-black-box reduction to circumvent the [IR89] oracle separation between Minicrypt and Cryptomania. To do so, we define and study a natural primitive called an oblivious interactive hash function (OIHF). An OIHF is a weakening of an oblivious pseudorandom function (OPRF), a widely studied object in applied cryptography [NR04, FIPR05]. We show the following:

(1) OIHFs are trivially constructible in the random oracle model (ROM), without any hardness assumption. This places OIHFs into Minicrypt.

(2) OIHFs are constructible from oblivious transfer (OT). This demonstrates that OIHFs exist from standard assumptions in non-idealized models.[1]

(3) The existence of an OIHF implies OT, via a non-black-box reduction. Therefore, OIHFs imply key agreement.

This blurs the line between Minicrypt and Cryptomania. Indeed, a common heuristic is to assume that a primitive constructible from only RO cannot imply public-key cryptography, but OIHFs demonstrate a concrete instance where this heuristic fails.

We emphasize that our results do *not* imply that OT/key agreement is possible from RO; our non-black-box reduction is well-defined only for OIHFs in non-oraclized models. This is to say that we do not propose an OT protocol that can be constructed from a cryptographic hash function. Rather, our result calls into question the value of the random oracle model as an idealization of Minicrypt and asks where the boundaries of Minicrypt lie.

1.1 Prior Work

The work of Damgård and Groth [DG03] pointed out that commitments in the universal composability (UC) framework of [Can01] imply key agreement, but are trivial in the programmable random oracle model (PROM). Specifically, [DG03] shows that one can construct a key agreement protocol by using the simulator from the security reduction from the UC commitment scheme, without making non-black-box use of any part of the protocol.

Thus, [DG03] may at first glance appear to achieve the same result as ours, but it does not. Indeed, [IR89] does not necessarily apply to simulation-based notions of RO, since simulation-based definitions can subtly change the model from what was considered by [IR89]. For the above case, such a construction points out the strength of the PROM, in particular that it gives a communication channel between the simulator and the adversary – namely, the programmed parts of the RO – that does not exist during any real-world protocol. Thus, the important observation of [DG03] says less about random oracles as an idealization of unstructured cryptographic primitives, and more about subtleties of simulation-based notions of security.

In contrast, our approach constructs OT from a definition that can be instantiated in the non-programming random oracle model (NPROM) [FLR+10]. Our OIHF protocol can be constructed in the ROM in such a way that [IR89] applies. Namely, [IR89] implies that OIHFs cannot yield public-key encryption in a black-box manner, and yet they imply OT anyway.

[1] Without this requirement, there would exist other trivial primitives that would "circumvent" [IR89], such as a concrete hash function H whose security is that it can always replace RO. Since such an H is impossible as shown in [CGH98], its existence would vacuously imply public-key cryptography.

1.2 Non-Black-Box Techniques

For the purposes of this work, a construction/reduction is non-black-box if it requires the existence of some primitive or assumption in the standard model, i.e. it requires that the primitive can be expressed with explicit code making no oracle accesses. Non-black-box techniques commonly appear as nested instances of cryptography, where one applies a cryptographic technique to the code of another. For instance, one might conduct a multi-party computation over the code of a pseudorandom function (PRF) to get an oblivious pseudorandom function (OPRF) [Yao86]. These types of non-black-box constructions can circumvent black-box barriers [BHH+25]. Despite formally requiring that the adversary have access to the embedded cryptography directly, many such reductions still make black-box use of the adversary, in the sense that a simulator or game interacts with the adversary in an input/output manner only.

There is a much rarer family of non-black-box techniques where the construction is black-box, but the security reduction's use of the adversary is not. To our knowledge, few works[2] use the adversary this way [Bar01, Bar02, BP15, BKP19, RV24, BT24]:

- In [Bar01, Bar02, BKP19], the authors use a non-black-box adversary to lift witness indistinguishable (WI) proof schemes to zero knowledge proof schemes. The authors achieve this by constructing simulators that use non-black-box details of the adversary to activate a trapdoor in the WI scheme.
- [BP15] is similar to [Bar01, Bar02, BKP19], except that [BP15] uses the fact that efficient unobfuscable functions exist to show that a simulator can extract certain information from the adversary's code.
- [RV24, BT24] give a win-win argument to build correlation robust hashes (CRHs) from multi-CRHs (MCRHs) by either concluding that a given MCRH is already a CRH, or by using the adversary that breaks the given MCRH to build a CRH.

Our approach also leverages a non-black-box adversary, though our proof structure is different.

As we will see, an OIHF is defined as an interactive protocol, but it is also natural to consider a *non-interactive* variant, and non-interactive OIHFs can be *trivially* achieved in the ROM. At the same time, it is not hard to show that standard-model non-interactive OIHFs *cannot exist*. When we construct our OT protocol from an OIHF, we prove security by showing that any adversary breaking our OT protocol must imply a non-interactive OIHF. But the adversary is a fixed algorithm without oracle access – it has access only to standard-model cryptography – so the fact that it yields a non-interactive OIHF is a contradiction.

1.3 Implications and Other Nuances.

Even though we do not posit a construction of OIHFs from any primitive in Minicrypt that existed prior to this work, we point out that some primitives live in Minicrypt in the same way that OIHFs do. Here are two examples:

- Collision resistant hash functions (CRHFs) are separated from one-way functions (OWFs) [Sim98], but are implied by RO or homomorphic encryption [IKO05], and
- Non-interactive zero-knowledge arguments (NIZKs) are only known in the ROM [FS87] or in the common reference string (CRS) model from public-key assumptions [FLS90, PS19, WWW25]. It is open whether CRS-based NIZKs exist from OWFs or other Minicrypt assumptions.

Both of the above primitives live in Minicrypt, but are not known from OWFs; however, they are known from various public-key assumptions. Even if one believes that OWFs should not yield public-key encryption, there are still primitives that live in Minicrypt only because they are derived from the ROM.

On the other hand, it could be that the ROM is too strong as a bound for Minicrypt, in that the "obliviousness" offered by virtue of being non-interactive may allow for strange but ultimately "harmless" oddities such as the result we present. In either case, there does not seem to be an obvious candidate to stand in as a formal upper-bound for Minicrypt, and defining Minicrypt as everything achievable from OWFs still throws many primitives, such as CRHFs and NIZKs, into limbo. If nothing else, we feel our results

[2]Other works cite and use the constructions introduced in the mentioned works, but we only include works that create a stand-alone construction.

emphasize that [IR89] does not provide an airtight seal between symmetric- and public-key cryptography, and that there exists an interesting opportunity to discover more about the complexity of Minicrypt.

For further discussion, see Section 5.

1.4 A Weaker Oblivious PRF and Uninstantiability

The main primitive we consider in this work is a natural weakening of oblivious pseudorandom functions (OPRFs). If one thinks of OPRFs as instantiating a functionality that allows a client to evaluate a PRF exactly once, then a natural question may be whether the OPRF definition could be weakened. We consider the case of an OPRF-like object that only has to achieve something that would be impossible non-interactively, such as correlation intractability [CGH98].

We call such a primitive an oblivious interactive hash function (OIHF), though when instantiated in the ROM, the protocol is not really interactive. Naturally, we can instantiate such a scheme in the standard model by noticing that an OPRF is strictly stronger, in the sense that an OPRF can replace the RO in a protocol at the expense of needing interaction. In general, a black-box construction of OIHFs from primitives inside of Minicrypt would not violate the black-box separation of [IR89].

We note that the above discussion covers points (1) and (2) from Section 1 w.r.t. OIHFs: the RO is an OIHF and OPRFs are OIHFs and can be achieved in a non-black-box way from OT.[3] The rest of the introduction is dedicated to illuminating how OIHFs achieve point (3).

1.5 Overview: Oblivious Transfer from OIHF

Consider a standard construction of OT from an OPRF. Namely, suppose we send the receiver two uniformly sampled strings x_0 and x_1, as well as two encryptions c_0 and c_1 of the following form,

$$c_b := m_b \oplus F_k(x_b)$$

for $b \in \{0, 1\}$, and where F is the PRF. Then the receiver can simply evaluate the OPRF scheme one time on x_b for their choice bit b to obtain $F_k(x_b)$ and then decrypt m_b. From OPRF security, message m_{1-b} is hidden.

Now, consider this exact protocol, except that we use an OIHF in the place of the OPRF. Naturally, any black-box reasoning must fail due to [IR89] since OIHF can be instantiated in the ROM. Consider a simpler game in which the adversary queries the OIHF at a uniform point x and then must guess the output of a second uniform point $\hat{x}$. If no adversary can do this, then this is called an oblivious unguessable function (see Definition 2), which importantly implies OT given standard techniques (see Section 4.3). In the ROM, the construction of an OIHF is simply a random oracle, in which case the adversary can easily win the above game by querying on the point $\hat{x}$ after the fact. This is unsurprising and consistent with [IR89].

However, now consider only those OIHF protocols that exist in the standard model. Suppose that there is an adversary that is still able to find the output $F_k(\hat{x})$ after observing only an interactive evaluation of $F_k(x)$. As we have just seen, the definition of OIHF does not prevent the adversary from guessing such a value with high probability; however, recall that any OIHF should retain the security properties of a random oracle. For simplicity, suppose the adversary guesses $F_k(\hat{x})$ with probability 1. Such an adversary is now an example of a standard model hash function which implements OIHF, which is also a correlation intractable function. But correlation intractability is uninstantiable, so such an adversary cannot exist.

More formally, our proof proceeds as follows. Any OIHF can (interactively) instantiate a correlation intractable function family [CGH98], and the existence of an adversary that can consistently guess the output of the OIHF would be a function family that agrees with the OIHF. This is in contradiction with the fact that OIHFs are a secure interactive correlation intractable "hash" family. Thus, no standard model adversary can win the oblivious unguessable function game against an OIHF, and therefore any *standard model* OIHF is an oblivious unguessable function, implying that OT exists.

[3] i.e. use OT to construct PRFs and MPC and then run the MPC over the PRF [FIPR05].

2 Preliminaries

Notation 1 (Common Notation)**.**

- *λ is a security parameter.*
- *ℓ is a key length, which should be at least λ and be polynomial in λ.*
- *$x \approx y$ denotes that ensembles x and y are computationally indistinguishable (in λ).*
- *$x \approx_O y$ denotes that ensembles x and y over oracles are computationally indistinguishable (in λ) when the oracles are given to a distinguisher.*
- *$\mathcal{O}$ denotes a random oracle.*
- *$a, b, \tau \leftarrow \langle A(x), B(y)\rangle$ denotes that interactive processes A and B run on inputs x and y respectively. As a result of their interaction, A outputs a and B outputs b; τ denotes the protocol transcript.*
- *$\langle x, y\rangle_\oplus$ denotes the inner-product over bitstrings x and y interpreted as vectors in $\mathbb{F}_2$.*
- *$r \leftarrow\!\!\$\ \{0,1\}^*$ denotes sampling a random tape of an appropriate polynomial length.*

Notation 2 (Ensemble Notation)**.** *As with probability environments, we will give the procedure for sampling variables used in an ensemble separated by a separator, such as in the following example.*

$$\left\{ F_k \;\middle|\; k \leftarrow Gen(1^\lambda) \right\}.$$

This ensemble is over an oracle F parameterized by k, where key k is generated by calling Gen. Note that whether the ensemble is over strings or over oracles will be clear from context, i.e. whether ensembles are being compared with $\approx$ or $\approx_O$.

Definition 1 (Oblivious Pseudorandom Function)**.** $\Pi = (Gen, F, C, S)$ *is an **oblivious pseudorandom function** (OPRF) for PPT procedure Gen, function $F : \{0,1\}^\ell \times \{0,1\}^\lambda \to \{0,1\}^\lambda$, and interactive PPT procedures C and S if the following properties are satisfied.*

- ***Correctness:*** *For all inputs $x \in \{0,1\}^\lambda$*

$$\Pr\left[y \neq F_k(x) \;\middle|\; \begin{array}{l} k \leftarrow Gen(1^\lambda) \\ y, \_, \_ \leftarrow \langle C(1^\lambda, x), S(1^\lambda, k)\rangle \end{array} \right] \leq \mathsf{negl}(\lambda).$$

- ***Client Privacy:*** *There exists a PPT simulator $\mathcal{S}_C$ such that for all inputs $x \in \{0,1\}^\lambda$ the following ensembles are indistinguishable (in λ),*

$$\left\{ (k, r, \tau) \;\middle|\; \begin{array}{l} k \leftarrow Gen(1^\lambda) \\ r \leftarrow\!\!\$\ \{0,1\}^* \\ \_, \_, \tau \leftarrow \langle C(1^\lambda, x), S(1^\lambda, k; r)\rangle \end{array} \right\} \approx \left\{ \mathcal{S}_C(1^\lambda) \right\}.$$

- *F **is a PRF:***

$$\left\{ F_k \;\middle|\; k \leftarrow Gen(1^\lambda) \right\} \approx_O \left\{ f \;\middle|\; f : \{0,1\}^\lambda \to \{0,1\}^\lambda \text{ is uniform} \right\}.$$

- ***Server Privacy:*** *There exists a PPT simulator $\mathcal{S}_S$ such that for all inputs $x \in \{0,1\}^\lambda$ and all keys $k \in Supp(Gen(1^\lambda))$, the following ensembles are indistinguishable:*

$$\left\{ (y, r, \tau) \;\middle|\; \begin{array}{l} r \leftarrow\!\!\$\ \{0,1\}^* \\ y, \_, \tau \leftarrow \langle C(1^\lambda, x; r), S(1^\lambda, k)\rangle \end{array} \right\} \approx \left\{ \mathcal{S}_S(1^\lambda, x, F_k(x)) \right\}.$$

Definition 2 ((One-Evaluation) Oblivious Weak Unguessable Function)**.** $\Pi = (Gen, F, C, S)$ *is an **oblivious weak unguessable function** (OWUGF) for PPT procedure Gen, function $F : \{0,1\}^\ell \times \{0,1\}^\lambda \to \{0,1\}^\lambda$, and interactive PPT procedures C and S if the following properties are satisfied.*

- ***Correctness:*** *Same as in Definition 1.*
- ***Client Privacy:*** *Same as in Definition 1.*
- ***(One-Evaluation) One-More Unguessability:***
For all PPT adversaries $\mathcal{A}$ and all $x \in \{0,1\}^\lambda$, the following probability is negligible:

$$\Pr\left[F_k(\hat{x}) = \hat{y} \;\middle|\; \begin{array}{l} k \leftarrow Gen(1^\lambda) \\ r \leftarrow\!\!\$ \{0,1\}^* \\ y, \_, \tau \leftarrow \langle C(1^\lambda, x; r), S(1^\lambda, k)\rangle \\ \hat{x} \leftarrow\!\!\$ \{0,1\}^\lambda \\ \hat{y} \leftarrow \mathcal{A}(1^\lambda, y, r, \tau, \hat{x}) \end{array}\right] \leq \mathsf{negl}(\lambda).$$

Definition 3 (Random Oblivious Transfer)**.** *$\Pi = (Rc, Sn)$ is a **random oblivious transfer** (rOT) protocol for PPT procedures Rc and Sn if the following properties are satisfied.*

- ***Correctness:*** *It should be that for any choice bit $b \in \{0,1\}$*

$$\Pr\left[m \neq m_b \;\middle|\; m, (m_0, m_1), \_ \leftarrow \langle Rc(1^\lambda, b), Sn(1^\lambda)\rangle\right] \leq \mathsf{negl}(\lambda).$$

- ***(Server) Obliviousness:*** *There exists a PPT simulator $\mathcal{S}_O$ such that for any choice bit $b \in \{0,1\}$, the following ensembles are indistinguishable:*

$$\left\{(m_0, m_1, r, \tau) \;\middle|\; \begin{array}{l} r \leftarrow\!\!\$ \{0,1\}^* \\ \_, (m_0, m_1), \tau \leftarrow \langle Rc(1^\lambda, b), Sn(1^\lambda; r)\rangle \end{array}\right\} \approx \left\{\mathcal{S}_O(1^\lambda)\right\}.$$

- ***(Client) Pseudorandomness:*** *For any choice bit $b \in \{0,1\}$ the following ensembles are indistinguishable:*

$$\left\{(m, m_{1-b}, r, \tau) \;\middle|\; \begin{array}{l} r \leftarrow\!\!\$ \{0,1\}^* \\ m, (m_0, m_1), \tau \leftarrow \langle Rc(1^\lambda, b; r), Sn(1^\lambda)\rangle \end{array}\right\}$$
$$\approx \left\{(m, u, r, \tau) \;\middle|\; \begin{array}{l} r \leftarrow\!\!\$ \{0,1\}^* \\ m, \_, \tau \leftarrow \langle Rc(1^\lambda, b; r), Sn(1^\lambda)\rangle \\ u \leftarrow\!\!\$ \{0,1\} \end{array}\right\}.$$

3 Oblivious Interactive Hash Functions

In the following, we will write the critical security portion of the definition – Correlation Intractability – in both the standard model and in the ROM for clarity; the ROM definition is the natural adaptation of the standard model definition, and the ROM definition is always satisfied by any standard model implementation, which simply ignores the provided random oracle.

Our Correlation Intractability definition is inspired by the usual notion of functional correlation intractability from [CGH98], but the following definition is suited to our proof.

Definition 4 (Oblivious Interactive Hash Function)**.** *A protocol $\Pi = (Gen, F, C, S)$ is an **oblivious interactive hash function** (OIHF) for PPT procedure Gen, function $F : \{0,1\}^\ell \times \{0,1\}^\lambda \to \{0,1\}^\lambda$, and interactive PPT procedures C and S if the following properties are satisfied.*

- ***Correctness:*** *Same as in Definition 1.*
- ***(Server) Obliviousness:*** *Same as Client Privacy in Definition 1.*
- ***(Variant) Correlation Intractability:*** *For all adversaries $\mathcal{A}$, all poly-size circuits with oracle access γ, and for all inputs $x \in \{0,1\}^\lambda$, the following probability is negligible:*

$$\Pr\left[\gamma^{\mathcal{R}}(m) = F_k(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \text{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^\lambda \\ k \leftarrow Gen(1^\lambda) \\ r \leftarrow \{0,1\}^* \\ y, \_, \tau \leftarrow \langle C(1^\lambda, x; r), S(1^\lambda, k)\rangle \\ m \leftarrow \mathcal{A}^{\mathcal{R}}(1^\lambda, y, r, \tau) \end{array}\right] \leq \mathsf{negl}(\lambda).$$

Definition 5 ((ROM) Oblivious Interactive Hash Function)**.** $\Pi^{\mathcal{O}} = (Gen, F^{\mathcal{O}}, C^{\mathcal{O}}, S^{\mathcal{O}})$ *is an* ***oblivious interactive hash function*** *(OIHF) for PPT procedure Gen, function* $F : \{0,1\}^{\ell} \times \{0,1\}^{\lambda} \to \{0,1\}^{\lambda}$*, and interactive PPT procedures* C *and* S *if the following properties are satisfied.*

- ***Correctness:*** *Same as in Definition 1, with provided random oracles for* F*,* C*, and* S*.*
- ***(Server) Obliviousness:*** *Same as Client Privacy in Definition 1, with provided random oracles for* F*,* C*, and* S*.*
- ***(Variant) Correlation Intractability:*** *For all adversaries* $\mathcal{A}$*, all poly-size circuits with oracle access* γ*, and for all inputs* $x \in \{0,1\}^{\lambda}$*, the following probability is negligible:*

$$\Pr\left[\gamma^{\mathcal{R}}(m) = F_k^{\mathcal{O}}(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \textit{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^{\lambda} \\ k \leftarrow Gen(1^{\lambda}) \\ r \leftarrow \{0,1\}^* \\ y, \_, \tau \leftarrow \left\langle C^{\mathcal{O}}(1^{\lambda}, x; r), S^{\mathcal{O}}(1^{\lambda}, k) \right\rangle \\ m \leftarrow \mathcal{A}^{\mathcal{O},\mathcal{R}}(1^{\lambda}, y, r, \tau) \end{array}\right] \leq \mathsf{negl}(\lambda).$$

Roughly, the above notion of correlation intractability states that an adversary cannot predict the output of F on some ideally-hashed input of its choice, even when the adversary has access to the ideal hash function.

4 Results

In this section we show the three properties of interest of OIHFs, namely:

(1) OIHFs are constructible in the ROM, without any hardness assumptions.

(2) OIHFs are constructible from OT.

(3) The existence of an OIHF implies rOT, via a non-black-box reduction.

We will cover the first two points in a single section, since they are almost immediate. The last point will be split into two sections: one that uses OIHFs to build a weak intermediate primitive, OWUGFs (Definition 2), and one that uses OWUGFs to build rOT (Definition 3).

4.1 Instantiating OIHF

Construction 1 (OIHF from RO)**.** *We give a construction of OIHF from RO.*

$Gen(1^{\lambda})$	$F_k^{\mathcal{O}}(x)$	$C^{\mathcal{O}}(1^{\lambda}, x)$	$S(1^{\lambda}, k)$
1 : $k \leftarrow\!\$\ \{0,1\}^{\lambda}$	*1 :* $y \leftarrow \mathcal{O}(k\|\|x)$	*1 :* **recv** k	*1 :* **send** k
2 : **return** k	*2 :* **return** y	*2 :* $y \leftarrow F_k^{\mathcal{O}}(x)$	*2 :* **return** $\perp$
		3 : **return** y	

Note that although S *is allowed access to RO, the above construction does not need it.*

Lemma 1 (OIHF is in Minicrypt)**.** *OIHF (Definition 5) can be constructed in the ROM.*

Proof. We show that Construction 1 is an OIHF. Correctness is immediate. Obliviousness is also immediate: the client never sends any messages to the server.

For Correlation Intractability, first recall the associated game.

$$\Pr\left[\gamma^{\mathcal{R}}(m) = F_k^{\mathcal{O}}(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \text{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^{\lambda} \\ k \leftarrow \text{Gen}(1^{\lambda}) \\ r \leftarrow \{0,1\}^* \\ y, \_, \tau \leftarrow \left\langle C^{\mathcal{O}}(1^{\lambda}, x; r), S^{\mathcal{O}}(1^{\lambda}, k) \right\rangle \\ m \leftarrow \mathcal{A}^{\mathcal{O},\mathcal{R}}(1^{\lambda}, y, r, \tau) \end{array}\right].$$

We see that $\mathcal{O}(k||\cdot)$ is equivalent to having a uniform function $f(\cdot)$, for a fixed key. In particular, an adversary that ignored queries not starting with k can do just as well. Thus, we can simplify to get the following:

$$\leq \Pr\left[\gamma^{\mathcal{R}}(m) = f(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \text{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^\lambda \\ \text{Sample uniform function } f : \{0,1\}^\lambda \to \{0,1\}^\lambda \\ m \leftarrow \mathcal{A}^{f,\mathcal{R}}(1^\lambda) \end{array}\right].$$

Then we consider that the probability that $\gamma^{\mathcal{R}}(m) = f(\mathcal{R}(m))$ is exponentially low. We start by pointing out that queries to f are useless if the output was not an output of $\mathcal{R}$. For any adversary, we can simply make all the $\mathcal{R}$ queries only, and then for the output m and queries to $\mathcal{R}$ we make all the queries $f(\mathcal{R}(m))$ and see if any of these agree with the corresponding $\gamma^{\mathcal{R}}(m)$ evaluation. If any of them match then submit that one otherwise output a random guess. This adversary does at least as well as the original one, since unpaired f queries do not help to determine which $\mathcal{R}$ queries would result in that query.

Next, we then find that the probability that $\gamma^{\mathcal{R}}(m) = f(\mathcal{R}(m))$ after any particular query m is negligible since f is a uniform function independent of γ and $\mathcal{R}$ is negligibly likely to have a collision. ☐

Lemma 2 (OPRFs are OIHFs)**.** *OPRFs (Definition 1) are OIHFs (Definition 4).*

Proof. We first note that OPRF Correctness and Client Privacy are the same as OIHF Correctness and Obliviousness, so these properties are satisfied trivially.

For Correlation Intractability, we start again with the associated game.

$$\Pr\left[\gamma^{\mathcal{R}}(m) = F_k(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \text{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^\lambda \\ k \leftarrow \text{Gen}(1^\lambda) \\ r \leftarrow \{0,1\}^* \\ y, \_, \tau \leftarrow \langle C(1^\lambda, x; r), S(1^\lambda, k)\rangle \\ m \leftarrow \mathcal{A}^{\mathcal{R}}(1^\lambda, y, r, \tau) \end{array}\right].$$

Then by Server Privacy of OPRFs we find that there exists a PPT simulator $\mathcal{S}_S$ such that

$$\leq \Pr\left[\gamma^{\mathcal{R}}(m) = F_k(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \text{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^\lambda \\ k \leftarrow \text{Gen}(1^\lambda) \\ m \leftarrow \mathcal{A}^{\mathcal{R}}(1^\lambda, \mathcal{S}_S(1^\lambda, x, F_k(x))) \end{array}\right] + \mathsf{negl}(\lambda).$$

Then by the Pseudorandomness of F, we have that

$$\leq \Pr\left[\gamma^{\mathcal{R}}(m) = f(\mathcal{R}(m)) \;\middle|\; \begin{array}{l} \text{Sample uniform function } \mathcal{R} : \{0,1\}^* \to \{0,1\}^\lambda \\ \text{Sample uniform function } f : \{0,1\}^\lambda \to \{0,1\}^\lambda \\ m \leftarrow \mathcal{A}^{\mathcal{R}}(1^\lambda, \mathcal{S}_S(1^\lambda, x, f(x))) \end{array}\right] + \mathsf{negl}(\lambda).$$

Then we can use the reasoning from Lemma 1 to see that this probability is also negligible, since the adversary does not even get to query f more than once. ☐

This is to say that standard model OT implies OIHFs, since OT implies OPRFs in a non-black-box manner [Yao86].

Corollary 1 (OT Implies OIHFs)**.** *Any standard-model OT protocol implies OIHFs exist.*

4.2 Non-Black-Box Reduction from OIHF to OWUGF

We will show that OIHFs are OWUGFs by showing that any adversary that would break One-More Unguessability (Definition 2) also breaks Correlation Intractability (Definition 4) of the OIHF. This is our main technical contribution, showing the key step into public-key cryptography from standard model OIHFs. As such, we will be using Definition 4, which does not include an RO, but is equivalent to Definition 5 if the protocol is not allowed to make RO queries.

Lemma 3 (Non-Ideal OIHFs are OWUGFs)**.** *Let Π be an OIHF (Definition 4) such that procedures C and S are instantiated in the standard model, i.e. the protocol does not make use of the RO. Then Π is also an OWUGF (Definition 2).*

Proof. First note that OWUGF Correctness and Client Privacy are satisfied by OIHF Correctness and Obliviousness directly.

Now we give a non-black-box reduction from OWUGF One-More Unguessability (Definition 2) to OIHF Correlation Intractability (Definition 4). We copy the One-More Unguessability game. For the sake of reaching a contradiction, suppose that there exists a PPT adversary $\mathcal{A}$ and input $x \in \{0,1\}^\lambda$ such that

$$\frac{1}{\mathsf{poly}(\lambda)} \leq \Pr\left[F_k(\hat{x}) = \hat{y} \;\middle|\; \begin{array}{l} k \leftarrow \mathrm{Gen}(1^\lambda) \\ r \leftarrow\!\$\ \{0,1\}^* \\ y, \_, \tau \leftarrow \langle C(1^\lambda, x; r), S(1^\lambda, k)\rangle \\ \hat{x} \leftarrow\!\$\ \{0,1\}^\lambda \\ \hat{y} \leftarrow \mathcal{A}(1^\lambda, y, r, \tau, \hat{x}) \end{array} \right].$$

This is to say that given one evaluation of the OIHF on some point x, the adversary $\mathcal{A}$ is able to compute $F_k(\hat{x})$ for a uniform $\hat{x}$ with some noticeable probability. Now we will use $\mathcal{A}$ to break the Correlation Intractability of the OIHF. We construct the adversary $\mathcal{A}'$.

$\gamma^{\mathcal{R}}(m)$	$\mathcal{A}'^{\mathcal{R}}(1^\lambda, y, r, \tau)$
$(1^\lambda, y, r, \tau) \leftarrow m$	**return** $(1^\lambda, y, r, \tau)$
return $\mathcal{A}(1^\lambda, y, r, \tau, \mathcal{R}(1^\lambda, y, r, \tau))$	

The challenge will then end with the comparison

$$\gamma^{\mathcal{R}}(1^\lambda, y, r, \tau) = \mathcal{A}(1^\lambda, y, r, \tau, \mathcal{R}(1^\lambda, y, r, \tau)) \stackrel{?}{=} F_k(\mathcal{R}(1^\lambda, y, r, \tau)).$$

Because $\mathcal{A}$ has the necessary, well-formed inputs it expects, and because the challenge $\hat{x}$ is uniform from the perspective of $\mathcal{A}$ – and the same between $\mathcal{A}$ and F_k – we find that $\mathcal{A}$ matches F_k with noticeable probability, and so adversary $\mathcal{A}'$ wins with noticeable probability, contradicting the Correlation Intractability of Π. Thus, no such adversary exists, and hence any standard-model OIHF is also a OWUGF. $\square$

4.3 rOT from OWUGF

We present a construction of rOT from OWUGF. The following is relatively standard, and is included for completeness.

The OWUGF scheme can be used in the standard OT from OPRF construction to get a random OT where the unchosen string is unguessable instead of pseudorandom. Then we can extract a pseudorandom bit from the unviewed unguessable string by applying Goldreich-Levin [GL89] to each string output of the OT to get random OT.

Construction 2 (rOT from OWUGFs)**.** *We give a construction of 1-bit random OT $\Pi_{rOT} = (Rc, Sn)$ from OWUGF $\Pi = (Gen, F, C, S)$.*

$Rc(1^\lambda, b)$	$Sn(1^\lambda)$
recv (x_0, x_1, a_0, a_1)	$k \leftarrow Gen(1^\lambda)$
interact $y_b \leftarrow C(1^\lambda, x_b)$	$x_0, x_1, a_0, a_1 \leftarrow\!\$\ \{0,1\}^\lambda$
return $\langle y_b, a_b\rangle_\oplus$	$y_0 \leftarrow F_k(x_0)$
	$y_1 \leftarrow F_k(x_1)$
	send (x_0, x_1, a_0, a_1)
	interact $S(1^\lambda, k)$
	return $(\langle y_0, a_0\rangle_\oplus, \langle y_1, a_1\rangle_\oplus)$

Note that if F_k is not efficient, then it can be approximated by simulating $\langle C(1^\lambda, \cdot), S(1^\lambda, k)\rangle$.

Lemma 4 (OWUGFs Imply rOT)**.** *rOT (Definition 3) can be constructed from OWUGFs (Definition 2).*

Proof. We show that Construction 2 is an rOT scheme in the selective semi-honest setting.

Correctness and Obliviousness are immediate from the Correctness and Client Privacy of OWUGFs.

For Pseudorandomness, we find that by the (One-Time) Unguessability of OWUGFs that y_{1-b} is unguessable. Then $\langle y_{1-b}, a_{1-b}\rangle_\oplus$ should be pseudorandom by Goldreich-Levin [GL89]. □

By combining with Lemma 3, we obtain the following.

Corollary 2 (Non-Ideal OIHFs Imply rOT)**.** *rOT (Definition 3) can be constructed from non-ideal OIHFs (Definition 4), i.e. procedures C and S are instantiated in the standard model.*

5 Discussion

We give some of our thoughts on various perspectives or approaches that future work may consider w.r.t. the results in this paper.

As a Construction. At its core, our result is constructive: given any standard model OIHF, we construct OT. In principal, future work may find pathways from other primitives to OT through OIHF, not necessarily starting from Minicrypt. For example, key agreement (KA) and public-key encryption (PKE) are protocols that are definitionally in Cryptomania, but are also formally separated from OT [GKM+00]. It could be that non-black-box use of these primitives is sufficient to construct OT, which would unify Cryptomania.

As a Lower-Bound. If one believes that OWFs cannot imply public-key cryptography, then a non-black-box construction of OT from some primitive Π might be seen as evidence that Π is not in Minicrypt. There are many more primitives in Minicrypt (as defined by the ROM) than there are primitives implied by OWFs, and in the event more primitives are shown to imply public-key cryptography, techniques similar to ours could become a new way to distinguish Minicrypt from Cryptomania.

Avenues for Separation. As discussed before, [IR89] provided a solid and seemingly airtight framework for categorizing primitives into either Minicrypt or Cryptomania. An alternative framework is to categorize primitives into (1) those that imply KA or (2) those that reduce to OWFs. However there could – and likely do – exist primitives which are stronger than OWFs but that do not imply public-key primitives. To avoid this problem, one might find other, weaker oracle separations between OIHFs and OWFs to erect more barriers between public-key cryptography and OWFs, but such oracle separations are liable to be circumvented to techniques similar to the ones presented in this paper.

The major weakness of oracle separations is that they say nothing about what is achievable by arbitrary reasoning. That said, one could potentially give a proof-theoretic separation between Minicrypt and Cryptomania. For example, one could show that the axiomatic system consisting of Zermelo-Frankel with Choice (ZFC), existence of OWFs, and existence of, say, KA is just as consistent as ZFC, existence of OWFs, and non-existence of KA. Formally, this would show that the (non-)existence of KA is up to axioms. Such a statement could be shown without implying anything about P versus NP, or even about whether OWFs exist, since in the event that one can show that OWFs do not exist, then both axiomatic systems are equally inconsistent.

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